

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Input 2 Transmitter Characterization	00AB	171	INT	R/W	0 = B TC 1 = E TC H 2 = E TC L 3 = J TC H 4 = J TC M 5 = J TC L 6 = K TC H 7 = K TC M 8 = K TC L 9 = NNM H 10 = NNM L 11 = Nicrosil H TC 12 = Nicrosil L TC 13 = R TC 14 = S TC 15 = T TC H 16 = T TC L 17 = W TC H 18 = W TC L 19 = 100 PT RTD 20 = 100 PT LO RTD 21 = 200 PT RTD 22 = 500 PT RTD 23 = Radiamatic RH 24 = Radiamatic RI 25 = Linear 26 = Square Root
Input 2 High Range Value	0023	035	FP	R/W	–999. to 9999. Engineering Units
Input 2 Low Range Value	0024	036	FP	R/W	–999 to 9999. Engineering Units
Input 2 Ratio	0025	037	FP	R/W	–20.00 to 20.00
Input 2 Bias	0026	038	FP	R/W	–999 to 9999. Engineering Units
Input 2 Filter	002B	043	FP	R/W	0 to 120 seconds
Input 2 Emmissivity	002C	044	FP	R/W	0.01 to 1.00

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Input 2 Burnout	00A5	0165	INT	R/W	0 = None 1 = Up 2 = Down 3 = No Failsafe

10.7.8 Control

Table 10-16 lists all the register addresses and ranges or selections for the function prompts in Set-up Group Control.

Table 10-16 Set-up Group – Control

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection	
	Hex	Decimal				
PV Source	0085	133	INT	R/W	0 = Input 1 1 = Input 2 2 = Input Algorithm 1	
Tuning Parameter Selection	00AC	172	INT	R/W	0 = One set only 1 = 2 sets keyboard selected 2 = 2 sets with PV automatic switchover 3 = 2 sets with setpoint (SP) automatic switchover	
Automatic Switchover Value (used with 172 selection 2 or 3)	0038	056	FP	R/W	Within the PV Range in engineering units	
Local Setpoint Source (Number of LSPs)	00AD	173	INT	R/W	0 = One Local Setpoint 1 = Two Local Setpoints 2 = Three Local Setpoints	
Power Up Mode Recall	0082	130	INT	R/W	Control Mode	Setpoint Mode
					0 = MAN	LSP
					1 = AUTO	LSP
					2 = AUTO	Last RSP
					3 = LAST	Last SP
					4 = LAST	Last Local SP
RSP Source	0083	131	INT	R/W	0 = None 1 = Input 2	
Setpoint Tracking	008A	138	INT	R/W	0 = None 1 = LSP = PV (when in Manual) 2 = LSP = RSP (when switched)	
Auto Bias	0089	137	INT	R/W	0 = Disable 1 = Enable	

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Control Setpoint High Limit	0007	007	FP	R/W	0 to 100% of PV (engineering units)
Control Setpoint Low Limit	0008	008	FP	R/W	0 to 100% of PV (engineering units)
Control Output Direction	0087	135	INT	R/W	0 = Direct 1 = Reverse
Output Rate Enable	009C	156	INT	R/W	0 = Disable 1 = Enable
Output Rate Up	006E	110	FP	R/W	0.00 to 9999% per minute
Output Rate Down	006F	111	FP	R/W	0.00 to 9999% per minute
High Output Limit	000E	014	FP	R/W	-5 to 105% of output
Low Output Limit	000F	015	FP	R/W	-5 to 105% of output
High Integral Limit	0010	016	FP	R/W	-5 to 105%
Low Integral Limit	0011	017	FP	R/W	-5 to 105%
Output Deadband for Time Duplex	004C	018	FP	R/W	-5 to +25.0%
Output Deadband for TPSC	0014	020	FP	R/W	0.5 to 5.0%
Output Drop Off Limit	0014	020	FP	R/W	-5 to 105%
Output Hysteresis	0013	019	FP	R/W	0.0 to 100.0% of PV
Failsafe Mode	00D5	213	INT	R/W	0 = Latching 1 = Non latching
Failsafe Output Level	0028	040	FP	R/W	0 to 100%
TPSC Power Output	00B7	183	INT	R/W	0 = Last 1 = Failsafe

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
TPSC Failsafe Output	00B8	184	INT	R/W	0 = Motor goes to closed position (0%) 1 = Motor goes to open position (100%)
Manual Output	0071	113	FP	R/W	0 to 100%
Automatic Output	0072	114	FP	R/W	0 to 100%
Proportional Band Units	0094	148	INT	R/W	0 = Gain 1 = Proportional band
Reset Units	0095	149	INT	R/W	0 = Minutes 1 = RPM

10.7.9 Options

Table 10-18 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Options.

Table 10-17 Set-up Group – Options

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Auxiliary Output *	0086	134	INT	R/W	0 = None 1 = Input 1 2 = Input 2 3 = PV 4 = Deviation 5 = Output 6 = Setpoint 7 = LSP 8 = RSP 9 = Input Algorithm 1
Low Scaling Factor	0031	049	FP	R/W	Within the range of the selected variable in ID 134
High Scaling Factor	0032	050	FP	R/W	Within the range of the selected variable in ID 134
Auxiliary Output Range	00EB	236	INT	R/W	0 = 4-20 mA 1 = 0-20 mA

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Digital Input #1	00BA	186	INT	R/W	0 = None 1 = To Manual 2 = To Local Setpoint #1 3 = To Local Setpoint #2 4 = To Local Setpoint #3 5 = To Direct Action 6 = To Hold Ramp 7 = To PID Set #2 8 = PV = Input 2 9 = Unused 10 = Rerun SPP Cycle 11 = To Run Ramp 12 = Reset SP Program 13 = Inhibit Reset 14 = To Manual/Failsafe Output 15 = Disable Keyboard 16 = To Automatic Output 17 = To Timer 18 = To Auto/Man Station 19 = Initiate Limit Cycle Tuning 20 = Setpoint Initialization (SP=Pv) 21 = Output Tracks Input 2 22 = Unused 23 = Unused 24 = To RSP 25 = Unused 26 = External Reset Feedback 27 = To Purge 28 = To Low Fire 29 = Manual Latching 30 = Unused 31 = Unused 32 = PV Hold

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Digital Input #1 Combinations	00BC	188	INT	R/W	0 = Disable 1 = +PID2 2 = +Direct 3 = +LSP2 4 = +Disable Accutune 5 = +LSP1 6 = +Run
Digital Input #2 *	00BB	187	INT	R/W	Same as Digital Input #1
Digital Input #2 Combinations	00BC	189	INT	R/W	0 = Disable 1 = +PID2 2 = +Direct 3 = +LSP2 4 = +Disable Accutune 5 = +LSP1 6 = +Run
* Auxiliary Output and Digital Input #2 are mutually exclusive.					

10.7.10 Communications

Table 10-18 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Communications.

Table 10-18 Set-up Group – Communications

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Shed Time	004F	79	INT	R/W	0 = No Shed 1 = 255 sample periods
Shed Mode and Output	00A2	162	INT	R/W	0 = Last Mode and Last Output 1 = Manual Mode, Last Output 2 = Manual Mode, Failsafe Output 3 = Automatic Mode
Shed Setpoint Recall	00A3	163	INT	R/W	0 = To Last Local Setpoint used 1 = CSP
Computer Setpoint Ratio	005A	90	FP	R/W	–20.00 to 20.00
Computer Setpoint Bias	005B	91	FP	R/W	–999 to 9999.
Communication Address	004D	77	FP	R/W	1 - 99
Communications Type	00E7	231	INT	R/W	0 = None 1 = Disable 2 = RS-485 Modbus 3 = Ethernet
IR Port Enable	00F1	241	INT	R/W	0 = Disable 1 = Enable
Baud Rate	00E8	232	INT	R/W	0 = 4800 1 = 9600 2 = 19200 3 = 38400
Transmit Delay	004E	78	FP	R/W	Response Delay in ms (1 to 500) +6ms
Floating Point Byte Order	00E9	233	INT	R/W	0 = Big Endian 1 = Big Endian Byte Swap 2 = Little Endian 3 = Little Endian Byte Swap
Shed Enable	00EA	234	INT	R/W	0 = Enable 1 = Disable
Shed Time	009A	154	INT	R/W	0 = No Shed 1 = 255 sample periods
Comm Data Units	00A1	161	INT	R/W	0 = Percent 1 = Engineering Units

10.7.11 Alarms

Table 10-19 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Alarms.

Table 10-19 Set-up Group – Alarms

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Alarm 1 Setpoint 1 Value	0009	009	FP	R/W	Within the range of selected parameter or PV span for deviation alarm
Alarm 1 Setpoint 2 Value	000A	010	FP	R/W	Within the range of selected parameter or PV span for deviation alarm
Alarm 2 Setpoint 1 Value	000B	011	FP	R/W	Within the range of selected parameter or PV span for deviation alarm
Alarm 2 Setpoint 2 Value	000C	012	FP	R/W	Within the range of selected parameter or PV span for deviation alarm
Alarm 1 Setpoint 1 Type	008C	140	INT	R/W	0 = None 1 = Input 1 2 = Input 2 3 = PV 4 = Deviation 5 = Output 6 = Alarm on Shed 7 = SP Event On 8 = SP Event Off 9 = Manual 10 = Remote Setpoint 11 = Failsafe 12 = PV Rate of Change 13 = Alarm on Digital Input 1 14 = Alarm on Digital Input 2 15 = Loop Break 16 = T/C Warning 17 = T/C Fail 18 = PV Hot Start

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Alarm 1 Setpoint 2 Type	008E	142	INT	R/W	Same as 140
Alarm 2 Setpoint 1 Type	0090	144	INT	R/W	Same as 140
Alarm 2 Setpoint 2 Type	0092	146	INT	R/W	Same as 140
Alarm 1 Setpoint 1 Event	008D	141	INT	R/W	0 = Low Alarm 1 = High Alarm
Alarm 1 Setpoint 2 Event	008F	143	INT	R/W	0 = Low Alarm 1 = High Alarm
Alarm 2 Setpoint 1 Event	0091	145	INT	R/W	0 = Low Alarm 1 = High Alarm
Alarm 2 Setpoint 2 Event	0093	147	INT	R/W	0 = Low Alarm 1 = High Alarm
Alarm Hysteresis	0029	041	FP	R/W	0.0 to 100% of output or span
Alarm Latching for Output 1	00C8	200	INT	R/W	0 = Non Latching 1 = Latching
Alarm States	00C9	201	INT	R/W	State = 0 = Not in Alarm State = 1 = In Alarm Bit 0 = Alarm 11 State Bit 1 = Alarm 12 State Bit 2 = Alarm 21 State Bit 3 = Alarm 22 State Event = 0 = Low Event = 1 = High Bit 4 = Alarm 11 Event Bit 5 = Alarm 12 Event Bit 6 = Alarm 21 Event Bit 7 = Alarm 22 Event
Alarm 1 Blocking	00CA	202	INT	R/W	0 = Disable 1 = Block 1 2 = Block 2 3 = Block 1 2

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Diagnostic Alarm	009A	154	INT	R/W	0 = Disable 1 = Alarm 1 2 = Alarm 2

10.7.12 Display

Table 10-20 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Display.

Table 10-20 Set-up Group – Display

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Decimal Point Location	009B	155	INT	R/W	0 = None – Fixed 1 = One – Floating decimal point to one 2 = Two – Floating decimal point to two 3 = Three – Floating decimal point to three
Temperature Units	0081	129	INT	R/W	0 = °F 1 = °C 2 = None
Power Frequency	00A6	166	INT	R/W	0 = 60 Hertz 1 = 50 Hertz
Language (Displays)	00C0	192	INT	R/W	0 = English 1 = French 2 = German 3 = Spanish 4 = Italian
Ratio Input 2 from Front Panel	00D0	208	INT	R/W	0 = Disable 1 = Enable

10.8 Modbus RTU Exception Codes

Introduction

When a master device sends a query to a slave device it expects a normal response. One of four possible events can occur from the master's query:

- *Slave device receives the query without a communication error and can handle the query normally.*
It returns a normal response.
- *Slave does not receive the query due to a communication error.*
No response is returned. The master program will eventually process a time-out condition for the query.
- *Slave receives the query but detects a communication error (parity, LRC or CRC).*
No response is returned. The master program will eventually process a time-out condition for the query.
- *Slave receives the query without a communication error but cannot handle it (i.e., request is to a non-existent coil or register).*
The slave will return with an exception response informing the master of the nature of the error (Illegal Data Address.)

The exception response message has two fields that differentiate it from a normal response:

Function Code Field:

In a normal response, the slave echoes the function code of the original query in the function code field of the response. All function codes have a most-significant bit (MSB) of 0 (their values are below 80 hex). In an exception response, the slave sets the MSB of the function code to 1. This makes the function code value in an exception response exactly 80 hex higher than the value would be for a normal response.

With the function code's MSB set, the master's application program can recognize the exception response and can examine the data field for the exception code.

Data Field:

In a normal response, the slave may return data or statistics in the data field. In an exception response, the slave returns an exception code in the data field. This defines the slave condition that caused the exception.